

Bingchang Wu

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EDUCATION

Georgia Institute of Technology (GPA: 4.0/4.0)

Atlanta, GA

Bachelor of Science in Computer Science; Threads: Intelligence and Systems & Architecture

Expected May 2028

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Organization & Programming, Discrete Mathematics, Linear Algebra, Objects & Design (Software Engineering)

EXPERIENCE

Software Engineering Intern

May 2026 – Aug. 2026

Saite Robotics | Python, FastAPI, PostgreSQL, ROS, LangChain, Chroma

- Deployed an internal diagnostic platform that converted ROS logs into technician-ready fault reports with centralized telemetry and reboot-safe recovery, reducing manual fault-triage time by ~80%.
- Grounded generated fault reports in maintenance guidance using a RAG pipeline, achieving 93% top-3 retrieval accuracy.

Undergraduate Research Assistant

July 2026 – Present

PAIR Lab, Georgia Tech | Python, PyTorch, Stable-Baselines3, MiniGrid

- Benchmarked 5 visual-navigation approaches and trained domain-randomized PPO across 3 layout families, achieving 100%, 98.4%, and 97.8% success across 42, 249, and 276 layout patterns after 40M steps.
- Built an LLM-as-judge pipeline with a 5-category spatial-reasoning taxonomy to classify 1,671 VLM reasoning steps across 40 navigation episodes.
- Increased RL training throughput 8× from ~200 to ~1600 FPS by eliminating redundant observation generation, parallelizing 8 environments, and removing unnecessary GPU overhead.

Section Leader

Apr. 2025 – Jun. 2025

Stanford University, Code in Place | Python

- Selected to teach Python to 15 students in Stanford's globally scaled Code in Place course serving 16,000+ learners.
- Led weekly live sections and office hours on control flow, functions, loops, and data structures, adapting explanations and debugging support to varied experience levels.

PROJECTS

Cue.ai | HackGT 13 | Chrome Extension, JavaScript, Python, FastAPI, WebGazer, ElevenLabs

September 2026

- Built a multimodal Chrome browser extension combining speech and eye tracking to search, compare, navigate, and add products hands-free, with deterministic purchase safeguards and passkey-gated checkout.
- Reduced simulated gaze noise 73% (70→19 px) and frame-to-frame jitter 89% (87→10 px), settling 400 px saccades in 260 ms by tuning One Euro filtering and dwell-based targeting.

csearch | Python, SQLite/FTS5, Vector DB (sqlite-vec), Sentence Transformers, tree-sitter

June 2026 – Present

- Built a fully local natural-language code search engine using AST/tree-sitter chunking, BM25, 768-dimensional code embeddings, and reciprocal rank fusion across lexical and vector retrieval.
- Achieved 91.7% recall@1 on a labeled query evaluation and 100.3 ms p50 warm-query latency across a 55,112-line, 2,132-chunk codebase using incremental SHA-256 indexing.

BuzzBoard | GT Web Dev Club | Next.js, TypeScript, Express, Firebase, Firestore

Jan. 2026 – May 2026

- Built features for a full-stack campus event-discovery platform supporting 500+ events, including real-time RSVP, friendship, and presence updates with optimistic client-side feedback.
- Designed proximity-aware feed ranking and schedule-conflict detection with Leaflet map visualization and ICS calendar import.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, C, C++, SQL, HTML/CSS

AI/ML: PyTorch, Stable-Baselines3, Reinforcement Learning (PPO), Vision-Language Models, RAG, Hugging Face, LangChain, LangGraph

Frameworks: Next.js, React, Express, FastAPI, Streamlit, Leaflet, Chrome Extensions

Developer Tools: Git, Docker, Linux (Ubuntu), ROS, Playwright

Databases & Cloud: PostgreSQL, SQLite, Chroma, Firebase, AWS, Vercel, Cloudflare